Algebra Review: Unit 8 — Quadratics

Station 1 of 4: Solving by Factoring and Square Roots

Name: _____ Date: _____

Solve each equation. Give exact answers; leave radicals in simplest form.

1. Solve: $x^2 + 5x + 6 = 0$

2. Solve: $x^2 - 7x + 12 = 0$

3. Solve: $x^2 - x - 20 = 0$

4. Solve: $2x^2 - 8x = 0$

5. Solve: $x^2 - 16 = 0$

6. Solve: $x^2 = 20$

7. Solve: $3x^2 - 12 = 0$

8. Solve: $(x - 3)^2 = 25$

9. Solve: $2x^2 + x - 6 = 0$

10. Solve: $(x + 1)^2 = 18$

Algebra Review: Unit 8 — Quadratics

Station 2 of 4: Completing the Square

Name: _____ Date: _____

Complete the square. Give exact answers.

1. What number completes the square? $x^2 + 8x + _____$

2. What number completes the square? $x^2 - 10x + _____$

3. What number completes the square? $x^2 + 3x + _____$

4. Solve by completing the square: $x^2 + 6x = 7$

5. Solve by completing the square: $x^2 - 8x + 7 = 0$

6. Solve by completing the square: $x^2 + 4x - 1 = 0$

7. Solve by completing the square: $x^2 - 2x - 5 = 0$

8. Solve by completing the square: $x^2 + x - 6 = 0$

9. Solve by completing the square: $2x^2 + 12x - 14 = 0$

10. Write $y = x^2 - 6x + 5$ in vertex form.

Algebra Review: Unit 8 — Quadratics

Station 3 of 4: The Quadratic Formula and the Discriminant

Name: _____ Date: _____

Use $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. *Give exact answers in simplest form.*

1. Find the discriminant of $2x^2 - 3x - 5 = 0$ and tell how many real solutions it has.

2. Find the discriminant of $x^2 + 4x + 4 = 0$ and tell how many real solutions it has.

3. Find the discriminant of $x^2 + 2x + 5 = 0$ and tell how many real solutions it has.

4. Solve: $x^2 - 3x - 10 = 0$

5. Solve: $2x^2 + 5x - 3 = 0$

6. Solve: $x^2 + 6x + 4 = 0$

7. Solve: $x^2 - 4x + 1 = 0$

8. Solve: $3x^2 - 2x - 2 = 0$

9. Solve: $4x^2 - 4x + 1 = 0$

10. Solve: $2x^2 - 6x + 3 = 0$

Algebra Review: Unit 8 — Quadratics

Station 4 of 4: Graphs of Quadratic Functions and Applications

Name: _____ Date: _____

Find the vertex, axis of symmetry, and other features. Include units in word problems. Diagrams are not drawn to scale.

1. For $y = x^2 - 4x + 3$, find the vertex, axis of symmetry, direction of opening, and y-intercept.

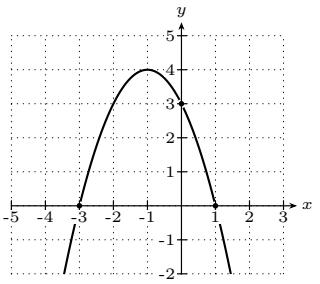
2. For $y = -x^2 + 6x - 5$, find the vertex, axis of symmetry, direction of opening, and y-intercept.

3. For $y = 2x^2 + 8x + 5$, find the vertex, axis of symmetry, direction of opening, and y-intercept.

4. Find the vertex and x-intercepts of $y = (x - 1)^2 - 4$.

5. Find the vertex and x-intercepts of $y = -2(x + 3)^2 + 8$. Does it open up or down?

6. The graph of a parabola is shown. Find (a) the vertex, (b) the axis of symmetry, (c) the x-intercepts, and (d) the y-intercept.



7. Find the vertex of $y = -3x^2 + 12x - 7$.

8. A ball is thrown so its height in feet is $h = -16t^2 + 32t + 48$, where t is in seconds. Find the maximum height and when it occurs.

9. For the ball in the previous problem, how long until it hits the ground?

10. A store's revenue (in dollars) is $R = -5p^2 + 100p$, where p is the price of an item. What price gives the maximum revenue, and what is that revenue?